

Uninterruptible Power System — Submittal Package

Submitted for review under Meridian Data Centre — Phase 1 (Nashik). This datasheet transmits the manufacturer's rated technical particulars for the Unit tagged UPS-02A for Engineer review and record.

Submittal No.	PMX-SUB-UPS02A-R3	Revision	R3
Project	Meridian Data Centre — Phase 1	Date	2026-07-20
Equipment Tag	UPS-02A	Spec Section	26 33 53 — Static UPS
Manufacturer	PowerMax Systems Pvt Ltd	Model	PMX-2000 Series
Vendor Ref.	PO-MDC-EL-0042	Status	Submitted — For Approval

1. Product Description

The PMX-2000 Series is a transformer-free, double-conversion (VFI-SS-111 per IEC 62040-3) three-phase uninterruptible power system intended for continuous-duty protection of critical data-centre load. The Unit continuously supplies the critical AC load from the inverter, with the rectifier drawing from the utility source to power the inverter and maintain the DC battery bank. On loss or degradation of the mains supply the Unit transfers to stored energy without interruption to the critical bus. A static bypass provides an alternate path for fault clearance and overload conditions, and a maintenance bypass permits concurrent servicing.

The Unit tagged UPS-02A is configured as a single stand-alone module suitable for parallel expansion. Ratings, tolerances and performance particulars are stated in Section 3. Values are manufacturer nameplate/type-test data at nominal conditions (400 V, 50 Hz, 25 °C) unless noted.

2. Configuration Summary

Attribute	Value
Topology	Double conversion on-line, transformer-free
Module arrangement	Single module (parallel-capable)
Cooling	Forced air, front-to-top
Battery technology (offered)	VRLA, 12 V blocks (Li-ion option available)
Enclosure	Free-standing floor cabinet, front access

3. Technical Specifications

3.1 Rated Output

Parameter	Rated Value	Unit	Condition / Note
Rated apparent power	2.1	MVA	At 0.9 pf, 25 °C
Rated active power	1.89	MW	Output pf 0.9 lagging
Output voltage (3-ph)	400	V	Selectable 380/400/415
Output frequency	50	Hz	±0.1% free-running
Output power factor	1.0	—	Full kVA = kW at unity
Output voltage THD (linear load)	1.5	%	100% linear load
Overload capability	125 / 10	% / min	Then transfer to bypass
Inverter → static-bypass transfer time	6	ms	Synchronised transfer

3.2 Rated Input (Rectifier)

Parameter	Rated Value	Unit	Condition / Note
Input voltage (3-ph)	400	V	3-phase + N
Input voltage tolerance	305 – 477	V	Without battery discharge
Input frequency range	45 – 65	Hz	—
Input current THD (iTHD)	4.5	%	At 100% linear load
Input power factor	0.99	—	At full load
Efficiency (double conversion)	96.5	%	At 100% load
Efficiency (economy / eco mode)	99.0	%	ECO mode enabled

3.3 Battery & Autonomy

Parameter	Rated Value	Unit	Condition / Note
Battery autonomy (backup time)	10	min	At 100% rated load
DC bus voltage (nominal)	480	V	40 × 12 V blocks
Recharge time to 90%	8	h	After full discharge
Battery type (offered)	VRLA	—	Li-ion optional

3.4 Physical & Environmental

Parameter	Rated Value	Unit	Condition / Note
Operating temperature	0 – 40	°C	Without derating to 40 °C
Relative humidity	0 – 95	%	Non-condensing
Acoustic noise @ 1 m	68	dB(A)	Full load
Ingress protection	IP20	—	Front access
Parallel capability	up to 4	units	N+1 redundancy supported
Dimensions (W×D×H)	1200×900×2000	mm	Single cabinet

Parameter	Rated Value	Unit	Condition / Note
Weight (excl. battery)	1150	kg	—

4. Standards & Compliance

The Unit is designed, type-tested and manufactured in accordance with the following standards. Classification per IEC 62040-3 is VFI-SS-111 (double conversion).

Standard	Scope
IEC 62040-1	UPS — General and safety requirements
IEC 62040-2	Electromagnetic compatibility (EMC) requirements
IEC 62040-3	Method of specifying performance and test — VFI-SS-111
IEC 60146-1-1	Semiconductor converters — general requirements
ISO 9001:2015	Quality management (manufacturing facility)

5. Certifications & Marks

CE marked in accordance with the applicable Low Voltage and EMC directives. Factory quality system certified to ISO 9001:2015. Type-test certificates and the routine (works) test report are available on request and will accompany the Factory Acceptance Test documentation.

6. Manufacturer Notes

- a) All ratings are stated at nominal reference conditions (400 V, 50 Hz, 25 °C) unless a condition is noted against the parameter.
- b) Values represent manufacturer nameplate and type-test data for the offered configuration.
- c) Site-specific derating, harmonic and short-circuit particulars will be confirmed at detailed design coordination.
- d) This submittal transmits manufacturer data only; conformity assessment against the Project Specification is to be determined by the Engineer during review.

Prepared by (Vendor)	Checked by (Vendor)	Engineer Review
R. Kulkarni Applications Engineering	S. Deshmukh QA / Product	Status / Date:

END OF SUBMITTAL — PMX-SUB-UPS02A-R3 This document contains manufacturer rated data for equipment tag UPS-02A. No conformity determination is made herein.